

PRIYAN SHAH

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EDUCATION

Purdue University

West Lafayette, IN

Bachelor of Science in Computer Engineering

Expected May 2027

- Relevant Coursework: Embedded Systems, Digital System Design, Data Structures and Algorithms, Computer Architecture, Software Engineering, Computer Security

EXPERIENCE

SoCET Team Member

Aug. 2025 – Present

SoCET Chip Implementation/PCB Research and Design - SoCET @ Purdue

West Lafayette, IN

- Contributed to 5+ PCB designs using KiCad and Altium for custom SoC, FPGA, and embedded hardware systems for testing and validation of SoCs
- Designed an FPGA interface PCB integrating external 3.3V SRAM, memory interfaces, and peripherals through an HSMC connector
- Developed hardware validation infrastructure for custom SoCs, supporting post-silicon testing, peripheral communication, and Embedded C firmware

Software and Design Intern

May 2025 – Aug. 2025; May 2026 – Aug. 2026

Spry Brands Inc.

Anderson, IN

- Automated 10+ customization, data-import, and production workflows, reducing customer data creation time by approximately 50%
- Supported 100+ clients through reusable website templates, backend workflows, and software components, allowing for smooth access of products
- Designed 300+ customizable customer-facing products using XML, PageFlex Studio, and Liff to add in variability for different designs

PROJECTS

LeRobot Automated Chess System | *Python, LeRobot, OpenCV, Robotics*

April - June 2026

- Integrated robotic arms, camera sensors, and a physical chessboard into an automated embedded robotics system for executing chess moves
- Developed an OpenCV vision pipeline to detect chess pieces and board positions, translating camera input into robotic control commands
- Implemented hardware-software communication and coordinated control loops for sensor processing, chess-state updates, and robotic actuation

Custom SoC Validation and FPGA Interface PCB | *KiCad, Altium, FPGA, SRAM*

August 2026 - Present

- Designed a custom PCB interfacing an FPGA development board with a custom SoC through a 120-pin socket and HSMC connector
- Integrated external SRAM, SPI Flash, EEPROM, UART, and I2C peripherals with supporting power regulation and signal routing
- Developed schematics and PCB layouts for 100+ signals, enabling hardware debugging, memory access, and post-silicon validation

RP2350 Embedded Game Boy System | *Embedded C, FreeRTOS, SPI, I2C, UART*

Nov. 2025 – Dec. 2025

- Developed 2,000+ lines of Embedded C firmware for an RP2350-based gaming system, integrating SPI, I2C, UART, and SD card peripherals
- Implemented FreeRTOS-based task scheduling and embedded control for game-state updates, hardware input, and 120x120 display rendering
- Validated peripheral communication, timing, and firmware functionality through 25+ test cases and 50+ debugging iterations

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Verilog, Java, JavaScript, MATLAB

Embedded & Firmware: Embedded C, FreeRTOS, RTOS Concepts, Microcontrollers, Interrupts, Peripheral Drivers, Hardware Debugging

Hardware & Interfaces: FPGA, PCB Design, SPI, I2C, UART, HSMC, SRAM, SD Card, Digital Logic, Analog Circuits

Software & Tools: Git, GitHub, Linux, KiCad, Altium, CAD, VS Code, Arduino, RP2350, OpenCV, XML

Languages: English (Fluent), Spanish (Conversational), Hindi (Conversational), Gujarati (Conversational)